

Nucleus

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surrounded by membrane called the nuclear membrane

- contains genetic information in the form of chromosomes

Function of Nucleus

- controls cell activities such as cell growth and repair of worn out parts
- essential for cell division (cells without nucleus like RBC cannot divide)

Chromosomes

- each chromosome $\Rightarrow$ long thread like structure $\Rightarrow$ found within nucleus (human cell has 46 chromosomes)
- made out of proteins and a compound called Deoxyribonucleic acid
- DNA $\Rightarrow$ stores hereditary information
- DNA contains information that cell needs for carrying out its activities.
- when cell is dividing $\Rightarrow$ chromosomes condense and shorten to form thick, rod like structures

Note: chromosomes are only visible during cell division due to their change in structure

Nuclear Membrane

$\Rightarrow$ separates contents of nucleus from rest of cytoplasm



